TECHNICAL REPORT

Survival Model Evaluation: flchain

Fitted with the survival-analysis-pipeline.

Source data	`datasets/flchain.csv.gz`
Rows	7,871 (27.5% with the ending observed, 72.5% censored)
Start dates	1995-01-01 to 2003-01-01
Evaluation	5 expanding-window temporal folds
Source of figures	`reports/flchain_demo/metrics.json`, regenerated by the command in section 9

1. Summary

This report evaluates two survival models fitted to `flchain.csv.gz`. It holds 7,871 rows, each observed from its start date. 27.5% have observed endings and 72.5% are censored, still running when observation stopped. Median observed duration is 4303 days. The models predict, from what was on file at the start date, how long each row survives.

The like-for-like comparison is the fold mean. Each of the 5 temporal folds fits both models on its own window, and the fold scores average. On concordance, the share of pairs a ranking orders correctly where 0.500 is a coin flip, XGBoost AFT scores 0.724 and the Cox proportional hazards baseline 0.756. The Cox baseline scores higher.

Pooled over 4,723 out-of-time test rows, the AFT model scores 0.742 (95% bootstrap interval 0.725 to 0.759). Section 5 explains how the two figures differ.

Both fitted models are saved. The bundle records the recommended model for scoring new rows.

Little of the pooled figure is `sex` group membership. Group means alone score 0.499, within-group comparisons 0.746. Section 5 gives the decomposition.

> This dataset has no known generating process, so no computable ceiling bounds these scores, and feature attributions cannot be checked against a true mechanism.

2. Motivation

From the run's notes, written by the analyst.

The serum free light chain cohort is one of the most widely taught survival datasets. It ships with R's survival package, it appears in the standard Python tutorials, and a reader who has studied survival analysis likely arrives already knowing roughly what a model scores on it. That makes it a useful public test. The pipeline's numbers can be checked against published ones, and where they differ, the difference has to be explainable.

The data is also a hard test. Nearly three quarters of the subjects were still alive when the study stopped watching, so a model here must learn to predict lifetimes from lifetimes it mostly never saw end. How each of the two models holds up under that much censoring is the run's main finding, and the interpretation section reports it.

3. Data

Durations are measured in days.

Left truncation, where a row was already running when the source's records begin, is a data fault. Its recorded start is not its true start and nothing marks it. Those rows must be excluded during preparation. Whether they were is recorded with the dataset.

From the run's notes, written by the analyst.

Each row is one subject in the Mayo Clinic serum free light chain study, observed from their blood sample until death or the end of follow-up. The features are the intake measurements: age at sampling, sex, the kappa and lambda free light chain concentrations, serum creatinine, and the assay's group decile. Only 28% of subjects died during follow-up. For everyone else, the data records how long they had lived so far, and nothing more.

The recorded start date is a year, with no month or day. Two subjects sampled in the same year therefore cannot be put in time order, and the limitations section says what that does to the folds.

Figure 1 shows Kaplan-Meier survival curves by `sex`, the coarsest structure in the outcome before any model is fitted.

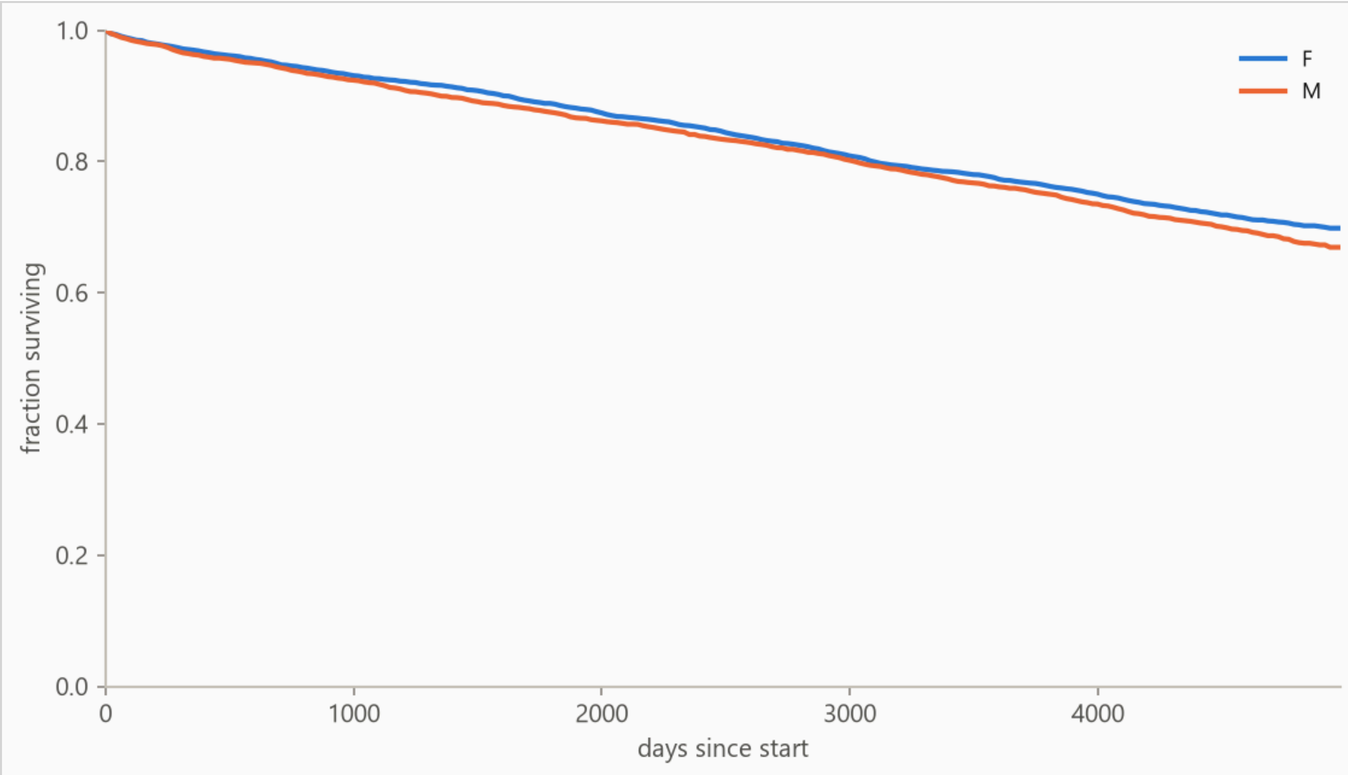

Figure 1. Kaplan-Meier survival curves by `sex`: the fraction of each group still running at each age, with censored rows counted for as long as they were observed. Groups beyond the seven most frequent are collapsed into (other) for this plot only.

4. Method

4.1 Model class

The target is a duration with incomplete observations, so the model is an accelerated failure time (AFT) model, XGBoost with its `survival:aft` objective. An observed ending enters as [t, t] and a censored row as [t, infinity), so every row contributes. The model predicts each row's median survival time in days, and a log-normal curve of fitted, shared width around that median gives the probability of surviving any horizon. A Cox proportional hazards baseline, the standard linear survival model, is fitted with lifelines' `CoxPHFitter` on the same features and answers whether the boosted model was necessary.

Numeric features pass through, missing values included (the Cox baseline gets train-window median imputation). Text columns are one-hot encoded with the vocabulary refit per training window, so a fold's features reflect only what was on file by its split date.

4.2 Temporal validation and label re-censoring

Evaluation uses 5 expanding-window folds ordered by start date: the earliest 40% of rows is burn-in, and each fold trains on every row started before its split date and tests on the next block (sizes in Table 2).

Training labels are re-censored at each split date. A row started long before a split may have died after it, and its label contains that future, so every post-split death is rewritten as a censoring at the split. Omitting this raises scores by importing the future, which makes it the most consequential detail in the pipeline. It is a standalone function (`cv.recensor`) with dedicated tests.

4.3 Selection and calibration

Hyperparameters are selected once, on the first fold's training window, by an inner temporal split scored on held-out censored log-likelihood. Concordance is scale-invariant and cannot see an unusable distribution width. The predictive scale, the reported curves' width, is measured on a temporal tail slice by a probe model that never trained on those rows, then carried over to the model refitted on the full window. The selected loss scale was 0.9 and the calibrated predictive scale 1.34. The probabilities this report states all use the calibrated width; the selected value is only the training loss's setting.

The methodology is validated separately against synthetic data with known ground truth.

5. Results

5.1 Discrimination

Table 1. Concordance index by model, pooled over 4,723 test rows with 95% percentile bootstrap intervals over 500 resamples. Higher is better, and 0.500 is a coin flip. Fold-mean rows score each of the 5 folds separately and average. The interval resamples test rows with the fitted models held fixed, so it measures scoring precision, not stability across history. The fold spread is the guide to that.

METHOD	CONCORDANCE (HARRELL'S C)	95% INTERVAL
XGBoost AFT (pooled)	**0.742**	**0.725 to 0.759**
XGBoost AFT (fold mean)	0.724	not resampled
Cox proportional hazards (fold mean)	0.756	not resampled

Cox risk scores are relative to each fold's own window, which is why fold-mean rows are the like-for-like comparison. The pooled row concatenates 5 separately fitted AFT models whose levels can differ, blurring cross-fold pairs. On this run it reads slightly high next to the fold mean.

The weakest window is fold 1, starting 1996-01-01, at 0.616. Any established cause is a dataset finding, and the run's notes are where it would appear.

The pooled concordance decomposes by `sex`: group means alone score 0.499, and comparisons restricted to same-group rows score 0.746 (pair-weighted across 2 qualifying groups). The within-group distance above 0.500 is row-level skill. The rest is group membership.

Figure 2 plots the per-fold concordance from Table 2.

Table 2. Per-fold results. Censoring is the share of each fold's test rows whose ending was not observed.

FOLD	SPLIT DATE	TRAIN N	TEST N	CENSORED	AFT	COX
1	1996-01-01	1,275	945	89%	0.616	0.649
2	1996-01-01	1,275	945	74%	0.786	0.847
3	1997-01-01	4,765	945	82%	0.655	0.680
4	1997-01-01	4,765	944	78%	0.798	0.809
5	1999-01-01	6,833	944	87%	0.764	0.797

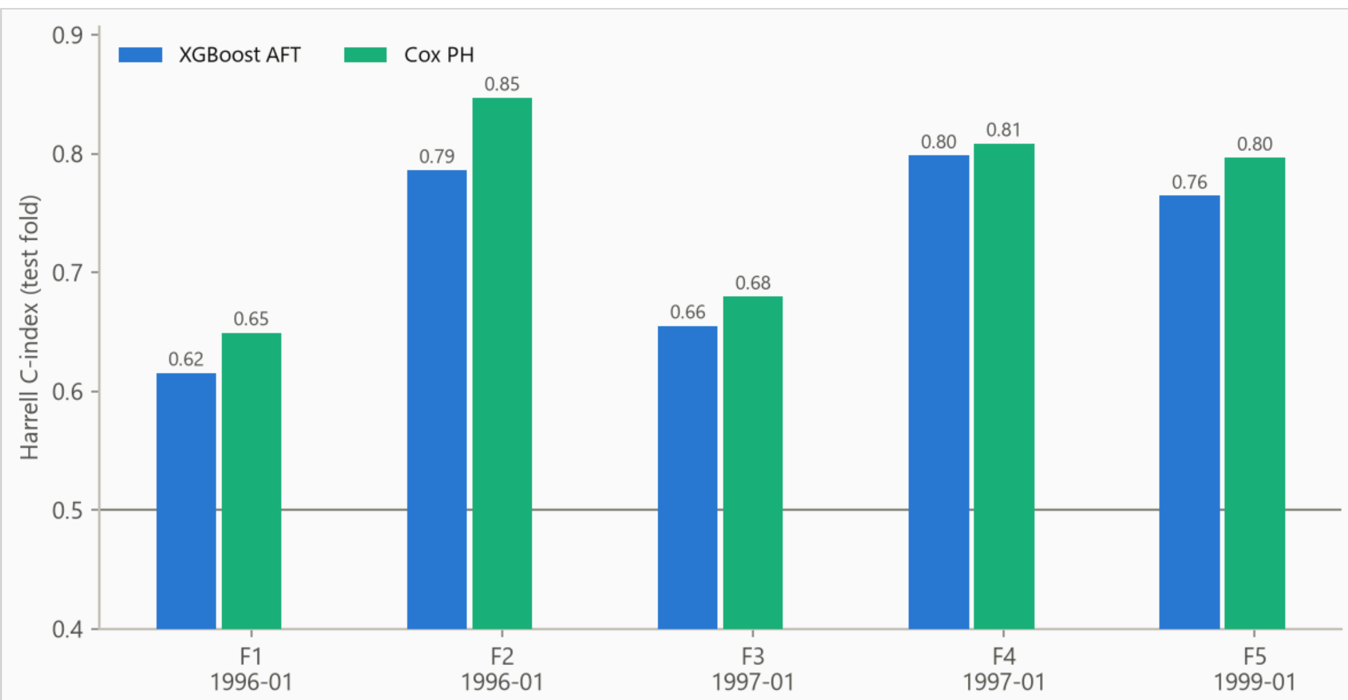

Figure 2. Concordance by fold, from 0.616 to 0.798. Folds differ in censoring mix, so cross-fold comparisons are indicative rather than exact.

5.2 Calibration

The Brier score, the mean squared error of a probability forecast (lower is better), is weighted by the inverse probability of censoring (IPCW) so censored rows do not bias it. The no-skill reference assigns every row the population's Kaplan-Meier marginal, the fraction surviving at each age with censored rows counted while observed.

Table 3. IPCW Brier score by horizon. Lower is better.

HORIZON	AFT	COX	NO-SKILL MARGINAL
1825 days	0.167	0.075	0.079
3650 days	0.321	0.144	0.138
4380 days	0.378	0.172	0.158

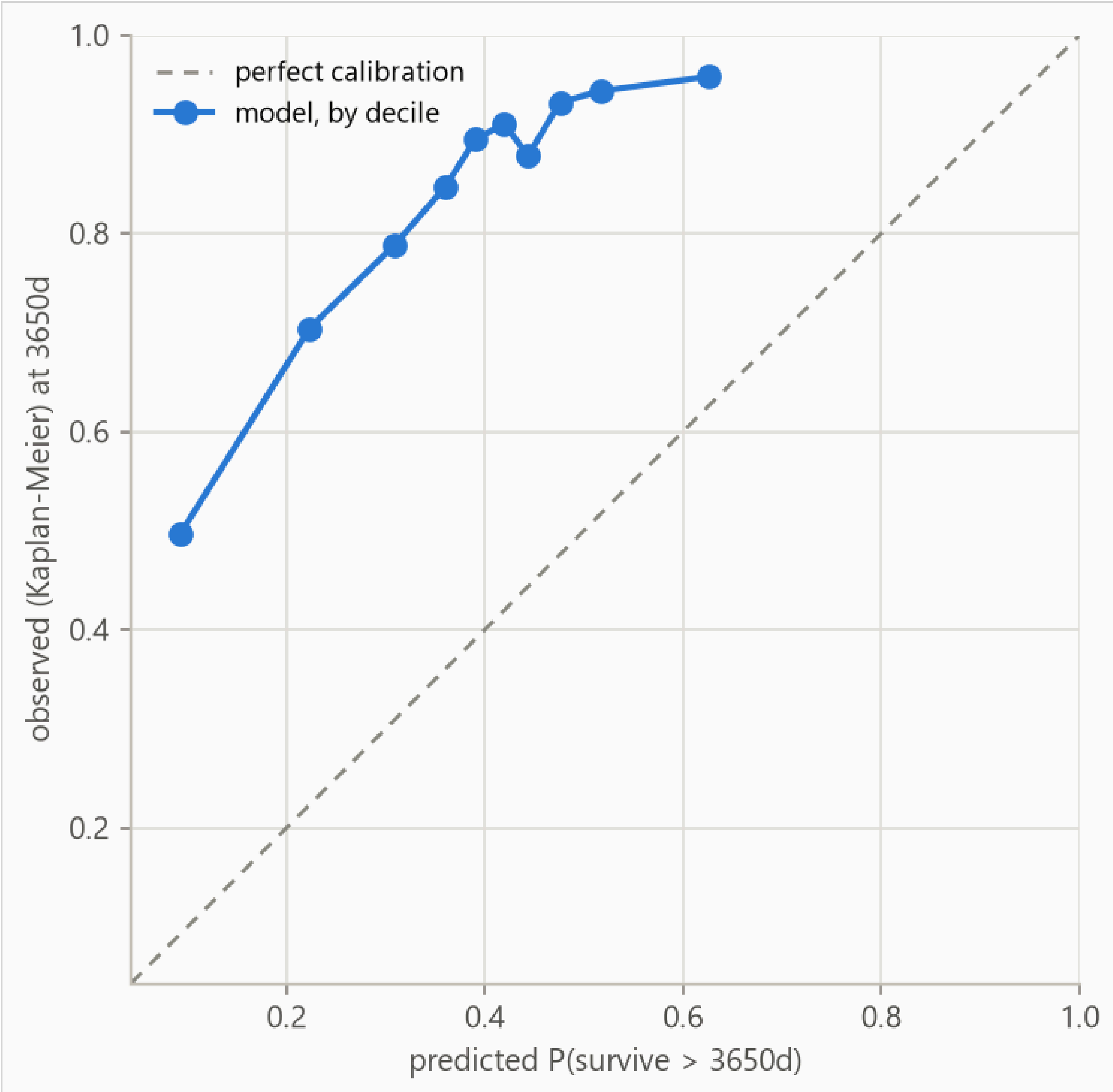

Figure 3. Predicted against observed survival at 3650 days, by predicted decile. Observed frequencies are Kaplan-Meier estimates within each bin, so censored rows contribute correctly. The largest deviation is 0.505 in decile 5.

Table 4. Decile calibration at 3650 days, the values plotted in Figure 3. Deciles are cut on predicted value, so tied predictions make the counts uneven. The observed value in each is a Kaplan-Meier estimate. When a decile's last observed row falls short of the 3650-day horizon the estimate carries its last value forward, so in small or heavily censored deciles the observed column is softer than its three decimals look.

DECILE	N	PREDICTED	OBSERVED (KM)	DEVIATION
1	473	0.093	0.497	+0.404
2	472	0.223	0.704	+0.481
3	472	0.309	0.789	+0.479
4	472	0.360	0.847	+0.487
5	474	0.391	0.895	+0.505
6	493	0.420	0.910	+0.490
7	454	0.444	0.878	+0.435
8	470	0.476	0.932	+0.456
9	477	0.516	0.944	+0.427
10	466	0.626	0.958	+0.332

6. What the model uses

Feature attributions (SHAP values, for SHapley Additive exPlanations) are computed on the log scale of survival time. A +0.3 attribution multiplies predicted survival by about 1.35, and negative values shorten it.

Attributions are descriptive and in-sample, computed on the final model's own training rows. Correlated features split credit by the model's internal choices as much as by the data, so directions are more trustworthy than magnitudes.

Figure 4 ranks features, Table 5 lists the top eight, Figure 5 shows the per-row spread, and Figure 6 traces attribution against value.

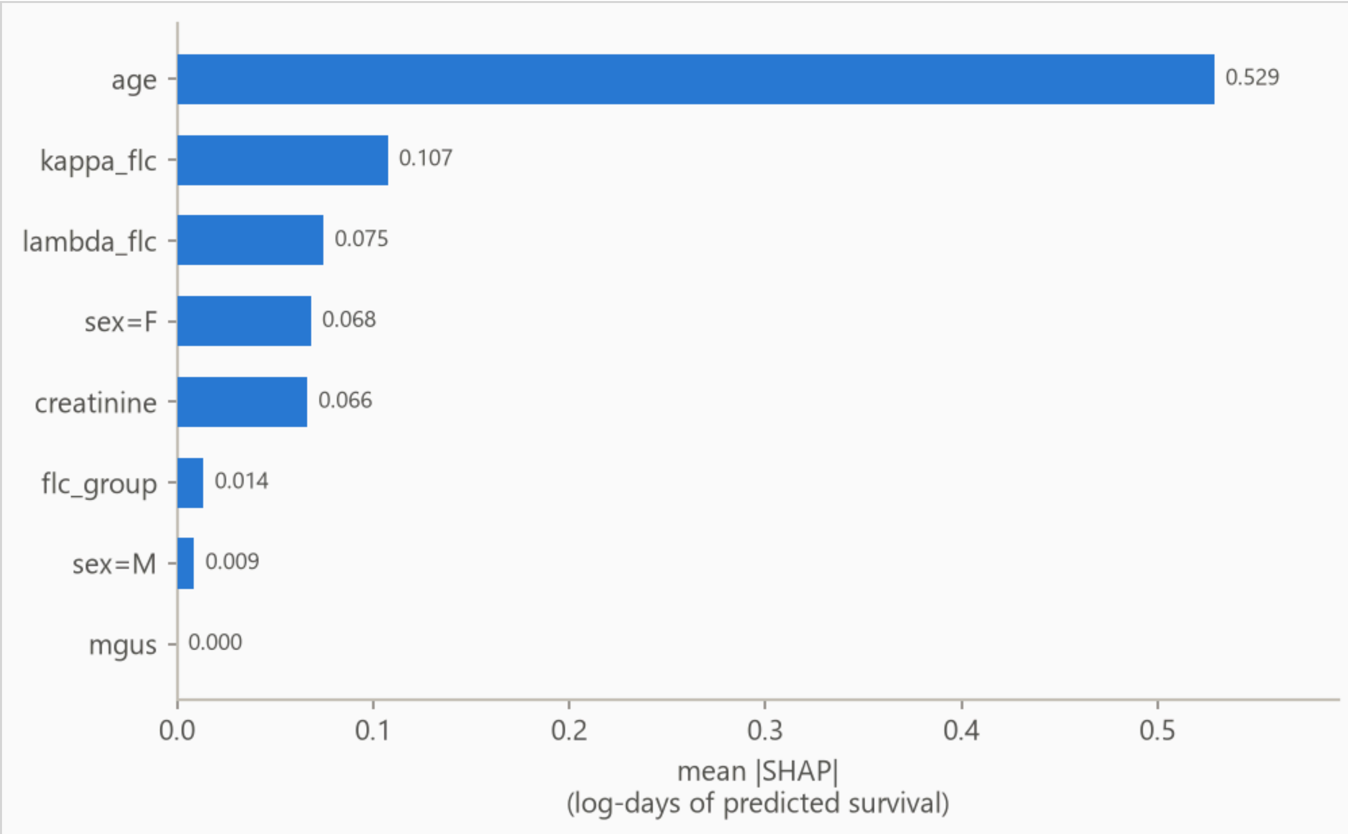

Figure 4. Mean absolute attribution by feature.

Table 5. Top eight features by mean absolute attribution.

FEATURE	MEAN |ATTRIBUTION|
age	0.529
kappa_flc	0.107
lambda_flc	0.075
sex=F	0.068
creatinine	0.066
flc_group	0.014
sex=M	0.009
mgus	0.000

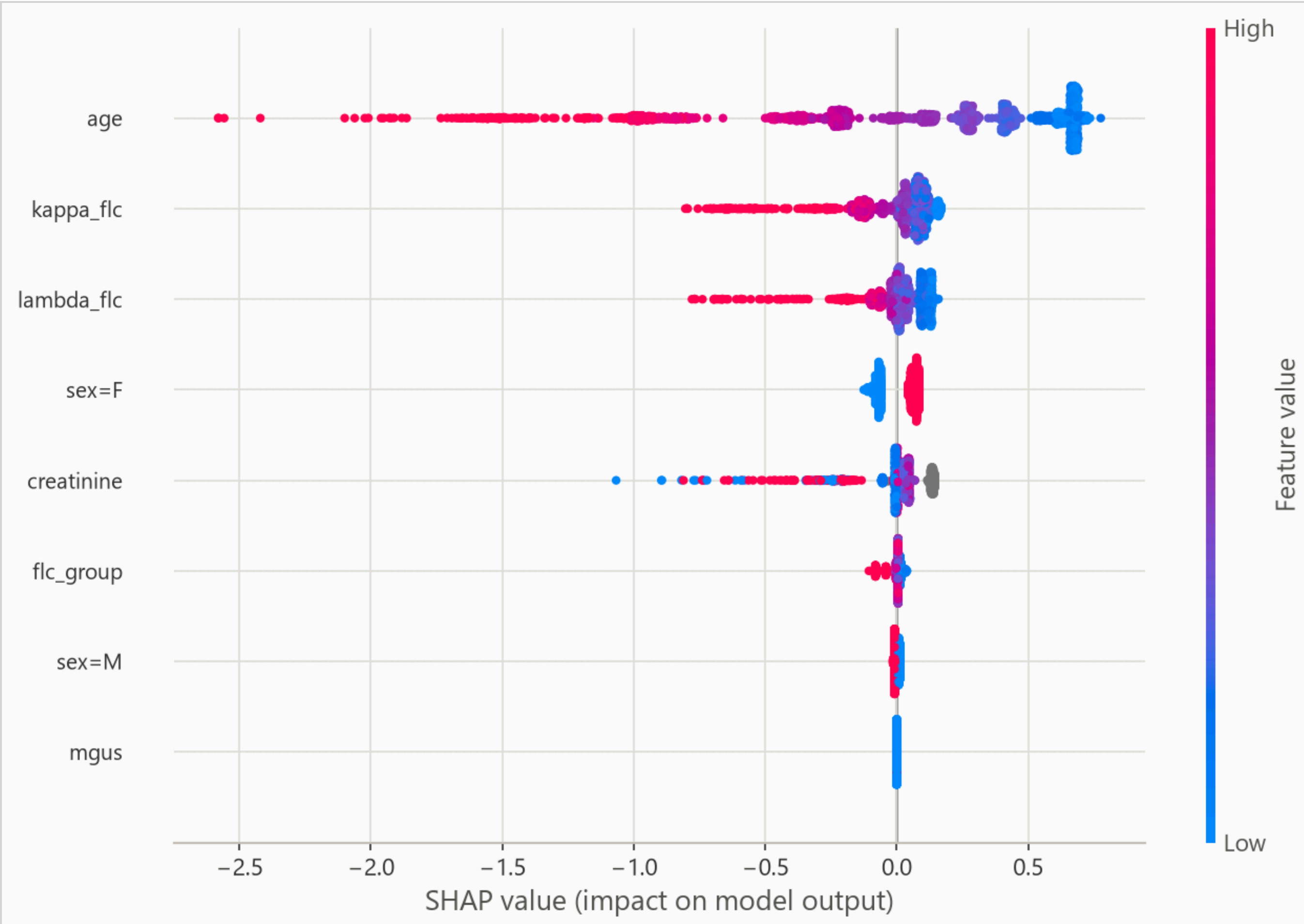

Figure 5. Per-row attributions across the explanation sample. For a yes/no feature, such as a one-hot category flag, the high (red) end simply means the row is in that category.

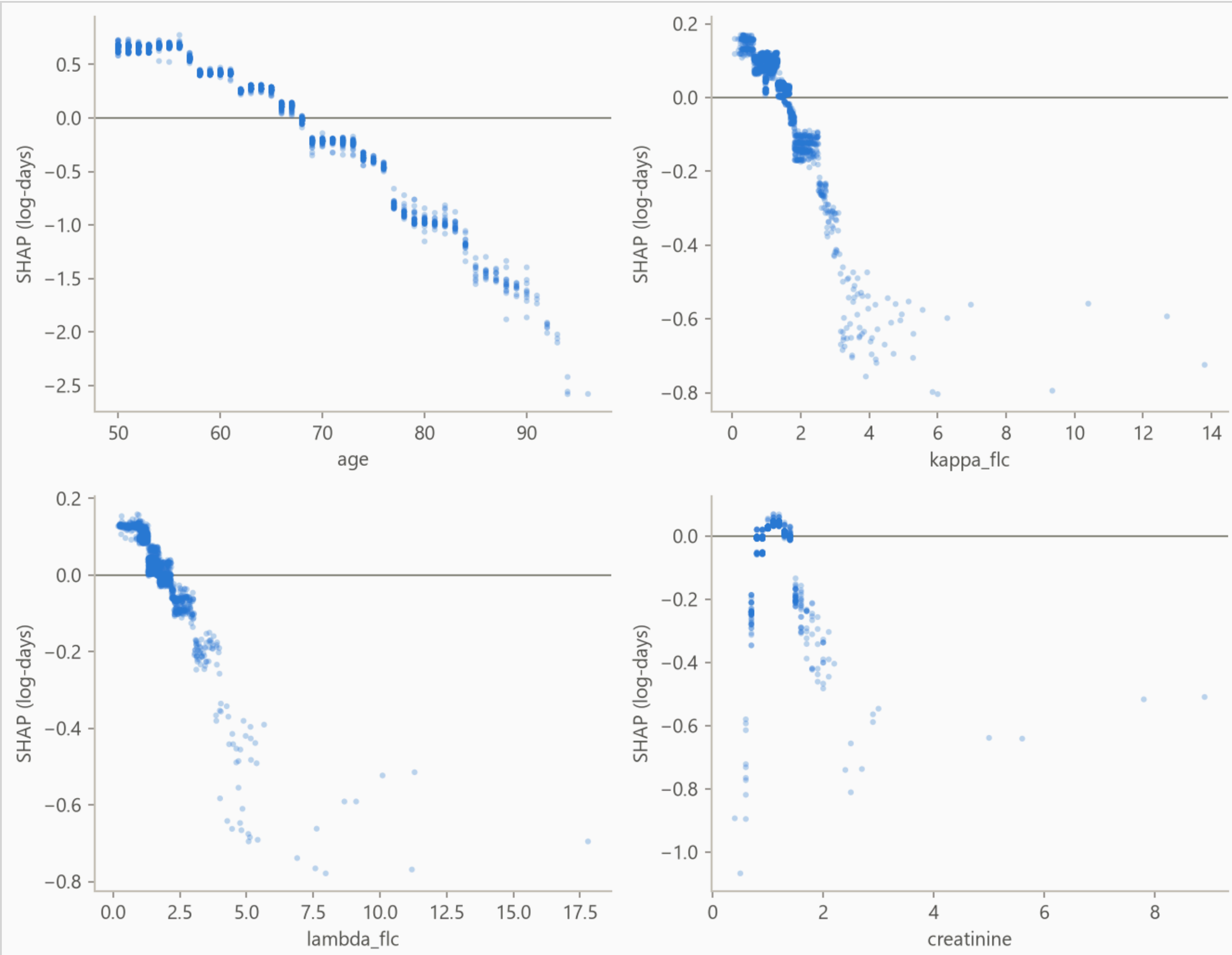

Figure 6. Attribution against feature value for the strongest numeric features. Category flags carry no shape and stay in the ranking above. A run short on numeric features fills the grid with flags instead.

On this data there is no known mechanism to check the ranking against, so read it as "what this model leaned on", not as importance in the world.

7. Interpretation

From the run's notes, written by the analyst.

How this compares to the standard exercise. This dataset is a teaching classic, and the usual exercise evaluates it with a random split. A random subset of people is hidden, the model trains on the rest, and its training pool therefore spans every year of the study. A published benchmark reports a concordance around 0.794 under that protocol. A random split suits the original research goal. A researcher asking whether the blood assay predicts mortality beyond age wants to know whether the relationship is real, the whole completed study is fair evidence for that, and the deliverable is the finding itself. Training only on the past suits the deployment goal. A clinic scoring each new patient at intake stands at a moment in time and knows only what has happened so far, and a test is only honest if it rehearses that position. This pipeline runs the second protocol, and its folds reconcile the two numbers. The late folds, whose training windows contain most of the study, score 0.809 and 0.797, matching the published figure. The earliest fold, trained before most deaths had happened, scores 0.649. The fold mean of 0.756 is the cost of training only on the past, and the fold column shows where that cost concentrates.

Why the boosted model's probabilities fail here. Only 28% of subjects died while the study watched. For everyone else, the data records only that they lived at least as long as the observation, so most true lifetimes lie beyond anything the model ever saw. The boosted model fits a fixed curve shape and places its guesses near the edge of what it observed, so every probability comes out too grim, and the Brier table above catches it losing badly to the no-skill forecast. The Cox baseline reads its curve off the data by counting who is still alive at each age, stays calibrated, and is the model this run's bundle recommends. This is why the pipeline fits both models and lets the data choose.

What the attribution shows. Age carries most of the ranking, and the two light-chain measurements add signal on top, which is the original study's own finding. The sex decomposition locates the model's contribution. Sex alone ranks survival at a coin flip, comparisons within a sex score 0.746, and the difference is signal the model found in the individual measurements.

8. Limitations

Absolute probabilities are not usable at 1825 days, 3650 days, 4380 days. The model loses to the no-skill forecast on the censoring-weighted Brier score there. Ranking and probability quality are separate, so use this model to order rows, not to act on absolute probabilities at those horizons.

Censoring may be informative. The evaluation assumes rows stop being observed for reasons unrelated to their risk. Early exits of rows about to end anyway bias survival estimates upward.

From the run's notes, written by the analyst.

The folds are only as sharp as the dates. Start dates here are years, so every subject sampled in the same year shares one date, and the folds cannot order them. A fold boundary that lands inside a year splits that year's subjects arbitrarily, so read the per-fold numbers loosely on this dataset.

Use the model to rank, not to read off long-horizon probabilities. This limitation is measured by the Brier rows in the results section, where lower is better. The boosted model's probabilities fail at every horizon, and the report says so above. The Cox model's probabilities hold up at five years and land just past the no-skill reference at ten years, a Brier score of 0.144 against the reference's 0.138. At long horizons the model still says who is at higher risk, and that is the use to put it to.

Every row shares one curve shape. The predictive scale is a single fitted number, so the model shifts a row's survival curve earlier or later but never reshapes it. Per-row width is future work.

Harrell's concordance is biased under heavy censoring, and the final fold is 87% censored. Read Table 2 with that in mind.

9. Reproducing this run

Python 3.11 or later. From the repository root:

```
python scripts/run_fit_evaluate.py --data datasets/flchain.csv.gz --name flchain --id-col subject_
python scripts/run_build_report.py --run reports/flchain_demo
```

Every number is read from the run's `metrics.json` at build time, and a figure the prose never cites fails the build. `requirements.txt` pins the exact versions the committed numbers came from.

Generated from `reports/flchain_demo/metrics.json` by `scripts/run_build_report.py --run`. 7,871 rows, 4,723 out-of-time test rows.